

Smart Metro Ticket Booking System

Project Development Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

1. User Acceptance Testing

1. Purpose

This document defines the User Acceptance Testing (UAT) approach for the metro digital ticketing solution built on ServiceNow. UAT confirms that the solution meets business requirements from the perspective of Passengers, Station Managers, the Metro Operations Team, and IT Admins before it is cleared for production go-live.

2. UAT Scope & Objectives

- Validate end-to-end ticket booking: selecting Source, Destination, Passenger Type, and Number of Tickets through the Service Catalog.
- Confirm automated fare calculation, including concessions for Student and Senior Citizen passenger types.
- Verify digital payment processing via UPI, credit/debit cards, and mobile wallets.
- Confirm QR code generation, delivery, and validation at AFC gates for both entry and exit.
- Validate Station Details Mapping, reporting dashboards, and role-based access for Station Managers, the Operations Team, and IT Admins.
- Confirm the solution works correctly across all channels: Mobile App, ServiceNow Portal, and WhatsApp.

3. UAT Process Flow

UAT follows a structured six-step process, from planning through stakeholder sign-off, with a retest loop for any defects found during execution.

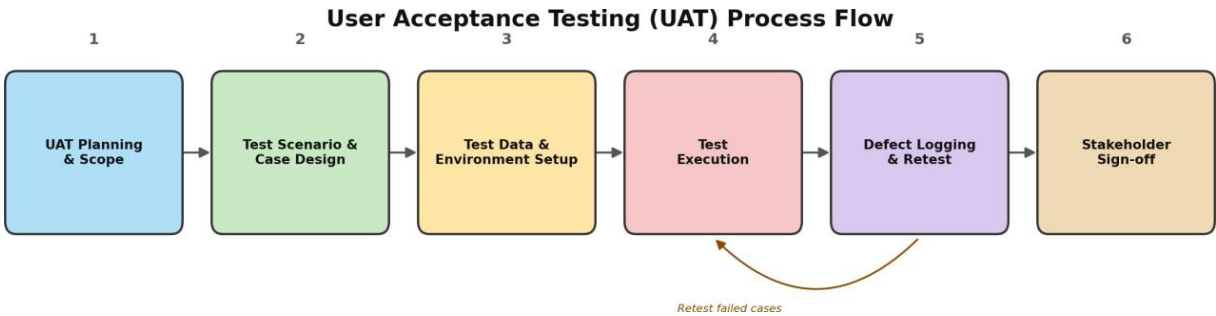


Figure — UAT Process Flow

4. UAT Test Scenarios

Scenario ID	Scenario Description	Related Milestone
UAT-SC-01	Passenger books a ticket for a single trip using the General passenger type.	Catalog Form Setup
UAT-SC-02	Passenger books multiple tickets in a single transaction.	Catalog Form Setup
UAT-SC-03	System applies the Student/Senior Citizen concession correctly.	Fare Logic Script
UAT-SC-04	Passenger completes payment via UPI, card, and wallet.	Fare Logic Script
UAT-SC-05	QR code is generated and delivered after successful payment.	QR Code Integration
UAT-SC-06	QR code is validated correctly at the AFC gate for entry and exit.	QR Code Integration
UAT-SC-07	Station Manager views real-time bookings for their assigned station.	Station Details Mapping

Scenario ID	Scenario Description	Related Milestone
UAT-SC-08	Passenger completes a booking via the WhatsApp channel.	UI Pages
UAT-SC-09	Operations Team updates fare/route configuration centrally.	Station Details Mapping

5. Sample UAT Test Cases

Test Case ID	Test Steps	Expected Result
TC-01	Open the catalog form; select Source, Destination, Passenger Type = General, Number of Tickets = 1; submit.	The fare is displayed correctly and the booking proceeds to payment.
TC-02	Repeat the booking with Passenger Type = Student.	The discounted fare is applied automatically per the concession policy.
TC-03	Complete payment using UPI on a successful test transaction.	Payment is confirmed and the QR code is generated within 5 seconds.
TC-04	Scan the generated QR at the AFC entry gate simulator.	The gate validates the QR as valid and logs entry; QR status updates to 'Scanned at Entry'.
TC-05	Attempt to reuse the same QR code for a second entry.	The system rejects the scan as invalid/reused and logs the attempt.
TC-06	Enter Source and Destination as the same station.	The system blocks submission with a validation message.

6. Defect Severity Classification

Defects found during UAT execution are classified by severity to help prioritize fixes. The chart below shows an illustrative distribution used for planning retest effort; actual figures will be tracked once execution begins.

Illustrative UAT Defect Severity Distribution

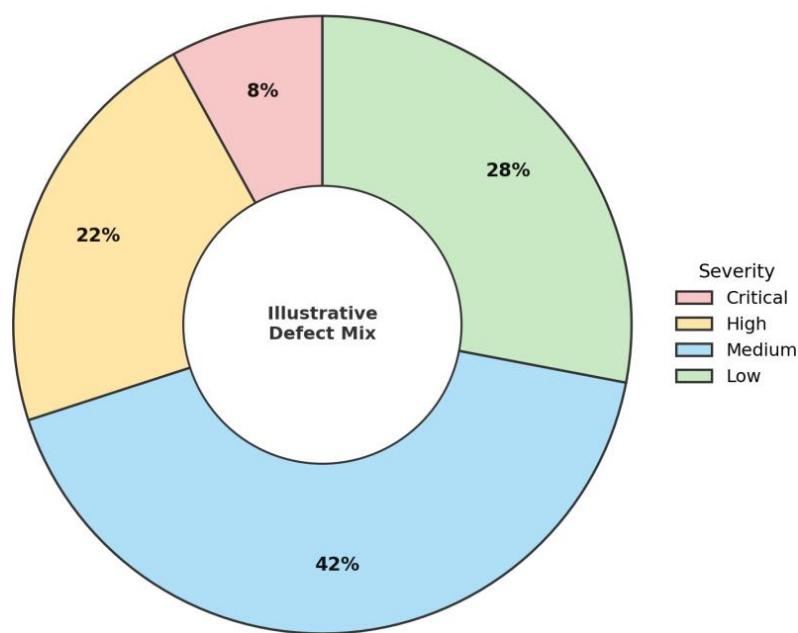


Figure — Illustrative UAT Defect Severity Distribution

7. UAT Team & Roles

Role	Responsibility
Passengers (Test Users)	Execute booking, payment, and gate scenarios from an end-user perspective.
Station Managers	Validate station-level dashboards, gate activity, and exception handling.

Role	Responsibility
Metro Operations Team	Validate fare/route configuration and reporting accuracy.
IT Admins	Support environment setup, defect triage, and integration verification.
QA / Test Lead	Own the UAT plan, coordinate execution, and track defects to closure.

8. Entry & Exit Criteria

8.1 Entry Criteria

- System Integration Testing (SIT) is complete with no open Critical/High defects.
- The UAT environment is configured with representative station and fare data.
- Test scenarios and test cases are reviewed and approved by stakeholders.
- UAT participants from all stakeholder groups are identified and available.

8.2 Exit Criteria

- All planned UAT test cases have been executed.
- No open Critical or High severity defects remain.
- Any open Medium/Low defects have documented workarounds and stakeholder agreement.
- Formal sign-off is obtained from the Passengers' representative, Station Managers, Operations Team, and IT Admins.

9. Sign-off

Once UAT is successfully completed against the exit criteria above, stakeholders will provide formal written sign-off, after which the solution proceeds to the Deployment milestone as defined in the Project Planning Phase document.